SALEH YAHYA

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EDUCATION

University of Illinois Urbana-Champaign Expected Dec 2027
M.S. in Computer Science
Quant at Illinois

Lewis University Graduating Dec 2026
B.S. in Computer Science
B.S. in Information Technology

EXPERIENCE

SoKat | *Software Engineer Intern* *July 2026 – Sep 2026*
- Engineered multi-agent LLM game platform in Python with crypto payment processing & strategy prompting for 20-round games.
- Refactored the monolith to distributed services on AWS EC2, SQS, CloudWatch, & Redis, scaling to 2k+ daily active users.
- Constructed idempotent BTC invoicing & Base USDC escrow settling $30k+, preventing duplicate invoices & replayed payouts.
- Integrated 6 LLM providers with OpenRouter & Fernet-encrypted keys, routing per model to shift inference cost to end users.

Paidwork | *Machine Learning Engineer Intern* *Sep 2025 – Nov 2025*
- Improved biometric verification reliability by implementing liveness detection using camera sensors with Python & PyTorch.
- Increased document verification accuracy by refactoring object detection and optical character recognition models across multiple international identification formats, improving text extraction and layout detection by 8% using TensorFlow.
- Strengthened fraud prevention by developing anti spoofing computer vision perception models using depth estimation and texture based analysis, reducing fraudulent authentication from static image and replay attacks by 37% to a user base of 25M.

OveoAI | *Software Engineer Intern* *May 2025 – Aug 2025*
- Migrated and transformed data pipelines from MySQL to PostgreSQL using Python, increasing database performance by 18%.
- Tuned denial model that automatically classifies denial reasons and generates corrected resubmission suggestions.
- Improved claim denial model accuracy by 12% by incorporating payer rules and coding compliance checks.

PUBLICATIONS & RESEARCH

Johns Hopkins University – Evaluating Frontier LLMs via Competitive Strategic Reasoning | First Author | SSRN Working Paper
- Designed a live benchmark where 5 frontier AI agents autonomously competed in Texas Hold'em under identical prompts & strategy.
- Instrumented 84 fields per decision to evaluate agent behavior, strategic reasoning, latency, token usage & inference cost.
- Measured AI strategic deception by comparing LLM agents' public table talk with private reasoning across 148 scored decisions.

Lewis University – MPMC Queue Performance Based Benchmarking | *C++, Python, SQL* | First Author | Preprint | GitHub
- Benchmarked MPMC queue performance by comparing mutex synchronization against lock-free atomic implementations in C++.
- Designed and executed multi-threaded simulations across 200K–6M operations measuring throughput, latency benchmarking.
- Identified system bottlenecks including memory safety, performance degradation, and lower throughput as data and threads grew.
- Transformed data into database with Python & SQL, visualized data with dashboards, enhanced research efficiency by 20%.

PROJECTS

Luminy – Financial Data Network & Analytics Platform | *Python,* Scikit-learn, *PostgreSQL, AWS, Redis, Nginx* | luminyy.com | GitHub
- Scaled distributed fintech platform to the **Top 200 in the App Store** Finance category using AWS EC2, CloudWatch, and CI/CD.
- Built asynchronous institutional ingestion workflows in Python, reducing financial management processing time by 40%.
- Reduced REST API response latency by 30% through Redis caching and optimized PostgreSQL indexing strategies.
- Developed ML financial analytics model using Scikit-learn for transaction classification & time series forecasting to 1k+ users.

Limit Orderbook Matching Engine | *C++, NoSQL, GTest* | GitHub
- Implemented a price, time priority limit order book in C++ processing microsecond level order matching for bid and ask levels.
- Designed custom priority queues and event driven logic supporting limit, market, and cancel orders enforcing FIFO execution.
- Simulated exchange style order flow and liquidity dynamics across thousands of order events under varying market conditions.

SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL
Frameworks & Tools: Kafka, Redis, WebSocket, REST API, PyTest, GTest, Matplotlib
Infrastructure: AWS, UNIX/Linux, CUDA, PostgreSQL, NoSQL, GCC, Nginx, Jenkins, Docker, Git, GitHub Actions
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Numpy, Pandas